



Bangladesh University of Professionals

Department of Computer Science and Engineering
Faculty of Science and Technology

Software Engineering LAB

CSE 3206

Assignment

Submitted By

Name	Roll
Abdul Hakim Shifat	23524202129
Mainul Islam	23524202047
SH Rony	23524202051
Shahrier Emon	23524202095

Level 2

The Level 2 DFD gives a more detailed view of the main processes shown in the Level 1 DFD. In this diagram, the core processes of the system are broken down into smaller sub-processes to show the internal data flow more clearly.

For this project, the Level 2 DFD mainly expands the Embed Secret Data process and the Extract Hidden Data process. The Authentication process and Admin / Monitoring process are kept as supporting processes because they are already clear in the Level 1 DFD.

1.1 Authentication

The Authentication process handles registration and login for the General User. The General User sends registration or login data to the system. The system reads from and writes to the User Database (D1) to store user records and verify login information. After checking the user information, the system returns an authentication result to the General User. If authentication is successful, a session token is provided for accessing the embedding and extraction processes.

1.2 Embed Secret Data

The Embed Secret Data process starts when an authenticated General User provides a cover file, secret data, and password. This process is divided into five sub-processes.

1.2.1 Receive Cover File and Secret Data:

The system receives the cover media file, secret text or file, and password from the General User.

1.2.2 Validate File and Password:

The system checks whether the provided file and password are valid. If the input is invalid, an error message is returned to the General User.

1.2.3 Embed Secret Data:

After successful validation, the system hides the secret data inside the cover file using steganography.

1.2.4 Generate Stego File:

The system generates the final stego file after embedding the secret data.

1.2.5 Save and Log Embed Operation:

The generated stego file is saved in the Stego File Store (D2), and the embedding operation is recorded in the Operation Log (D3). The stego file is also made available for download by the General User.

1.3 Extract Hidden Data

The Extract Hidden Data process starts when an authenticated General User provides a stego file and password. This process is divided into five sub-processes.

1.3.1 Receive Stego File and Password:

The system receives the stego file and password from the General User.

1.3.2 Validate Password:

The system checks the provided password and may read the stored stego file information from the Stego File Store (D2). If the password is wrong, the process returns an access denied result.

1.3.3 Extract Hidden Data:

If the password is correct, the system extracts the hidden data from the stego file.

1.3.4 Return Result:

The system returns either the extracted data or an access denied message to the General User.

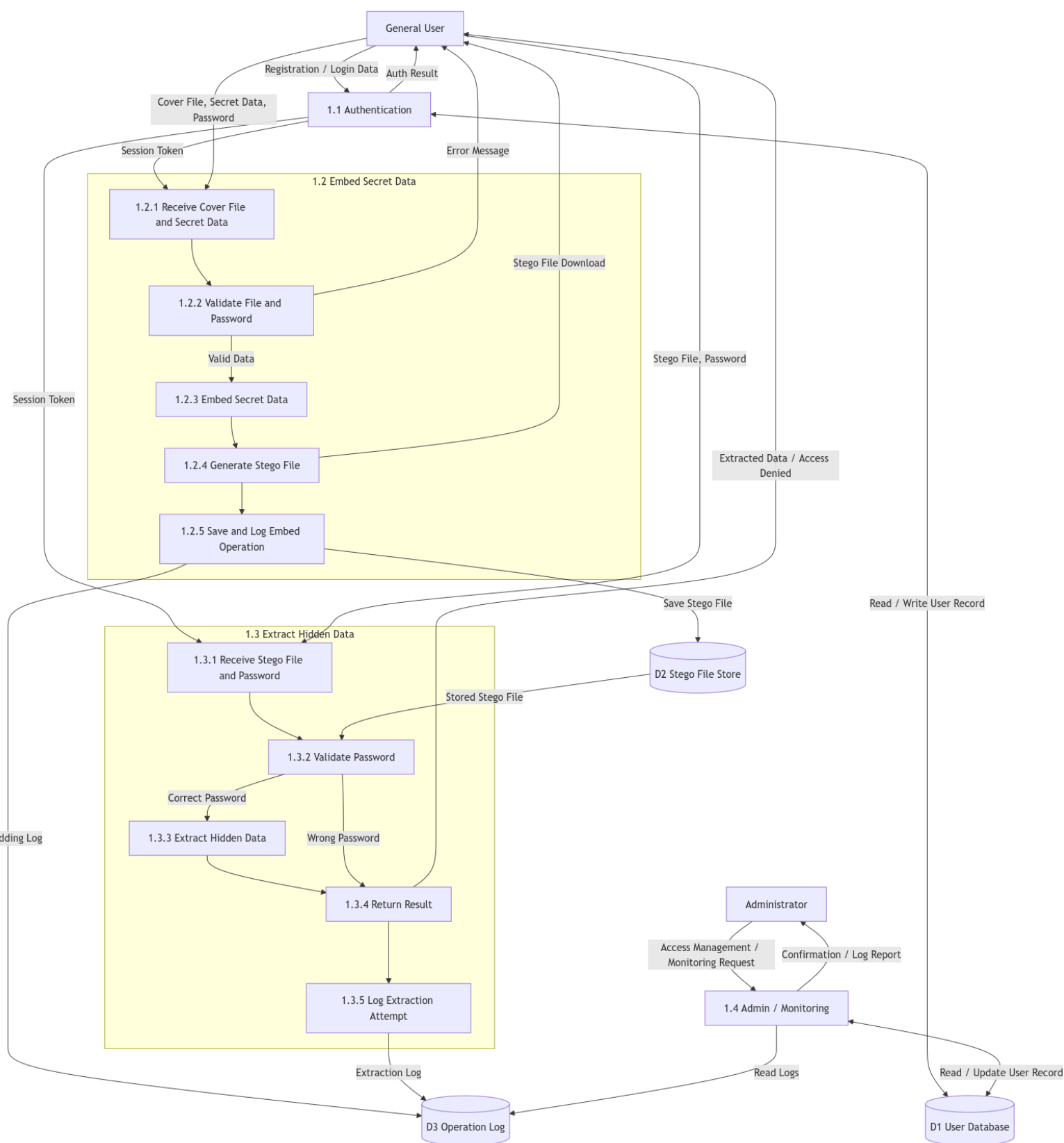
1.3.5 Log Extraction Attempt:

The system records the extraction attempt in the Operation Log (D3), whether the attempt is successful or unsuccessful.

1.4 Admin / Monitoring

The Admin / Monitoring process allows the Administrator to manage user access and monitor system activity. The Administrator sends an access management or monitoring request to the system. The system reads or updates user records in the User Database (D1), reads logs from the Operation Log (D3), and returns a confirmation message or log report to the Administrator.

Overall, the Level 2 DFD shows how data moves inside the main functions of the Secure Information Hiding System Using Steganography. It shows the flow of registration data, session tokens, cover files, secret data, passwords, stego files, extracted data, user records, and operation logs between the General User, Administrator, system processes, and data stores.



Database Schema

The proposed database schema contains three main tables:

1. **Users**
2. **StegoFiles**
3. **OperationLogs**

These tables are directly related to the data stores shown in the DFD. The **Users** table represents the User Database, the **StegoFiles** table represents the Stego File Store, and the **OperationLogs** table represents the Operation Log.

1. Users Table

The **Users** table stores information about all users of the system. It contains records for both General Users and Administrators. This table is used during registration, login, and admin access management.

Field Name	Data Type	Key	Description
user_id	INT	Primary Key	Unique identification number for each user
username	VARCHAR(100)		Username used for login
email	VARCHAR(150)		Email address of the user
password_hash	VARCHAR(255)		Encrypted or hashed password of the user
role	VARCHAR(20)		Defines whether the user is a General User or Administrator

Field Name	Data Type	Key	Description
status	VARCHAR(20)		Shows whether the user account is active, inactive, or blocked
created_at	DATETIME		Date and time when the user account was created

Description

When a new user registers, the system stores the user information in the Users table. During login, the system checks the provided username and password with the stored user records. The administrator can also read or update user records through the Admin / Monitoring process.

2. StegoFiles Table

The **StegoFiles** table stores information about generated stego files. A stego file is created after the system successfully embeds secret data into a cover media file.

Table Name: StegoFiles

Field Name	Data Type	Key	Description
stego_file_id	INT	Primary Key	Unique identification number for each stego file
user_id	INT	Foreign Key	ID of the user who created the stego file
cover_file_name	VARCHAR(255)		Name of the original cover media file
stego_file_name	VARCHAR(255)		Name of the generated stego file
file_type	VARCHAR(50)		Type or format of the file

Field Name	Data Type	Key	Description
file_size	BIGINT		Size of the generated stego file
storage_path	VARCHAR(255)		Location where the stego file is stored
password_protected	BOOLEAN		Shows whether password protection was used
created_at	DATETIME		Date and time when the stego file was created

Description

When a General User provides a cover file, secret data, and password, the system validates the input and embeds the secret data into the cover file. After successful embedding, the generated stego file information is saved in the **StegoFiles** table. This table helps the system keep records of files created by users.

3. OperationLogs Table

The **OperationLogs** table stores records of important system activities. It records embedding operations, extraction attempts, login activities, and admin monitoring activities.

Table Name: OperationLogs

Field Name	Data Type	Key	Description
log_id	INT	Primary Key	Unique identification number for each log record
user_id	INT	Foreign Key	ID of the user who performed the operation
stego_file_id	INT	Foreign Key	ID of the related stego file, if applicable

Field Name	Data Type	Key	Description
operation_type	VARCHAR(50)		Type of operation, such as Embed, Extract, Login, or Admin Update
operation_status	VARCHAR(50)		Status of the operation, such as Success, Failed, or Access Denied
message	TEXT		Short description of the operation
created_at	DATETIME		Date and time when the operation was performed

Description

Whenever a user performs an embedding or extraction operation, the system records the activity in the **OperationLogs** table. Successful and unsuccessful extraction attempts are also stored in this table. The Administrator can read these logs to monitor system activity.

Relationships Between Tables

The tables are connected using primary keys and foreign keys.

Relationship	Description
Users to StegoFiles	One user can create many stego files
Users to OperationLogs	One user can perform many operations
StegoFiles to OperationLogs	One stego file can be related to many operation logs

Relationship Explanation

The **Users** table has a one-to-many relationship with the **StegoFiles** table because one user can generate multiple stego files. The **Users** table also has a one-to-many relationship with the **OperationLogs** table because each user can perform many operations such as login, embedding, extraction, or admin actions. The **StegoFiles** table has a one-to-many relationship with the **OperationLogs** table because one

stego file may be involved in different operations, such as creation during embedding and later use during extraction.

